Mohamed Ali LAWINI

Embedded Software Engineer — C/C++ · Embedded Linux · STM32 / ARM · Automotive

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PROFILE

Embedded software engineer with **5 years of experience** at ACTIA Engineering Services, delivering projects for **STMicroelectronics** and **Renault**. I write C and C++ across the stack — from CAN firmware at **Linux kernel/driver** level and C++17 system services, to STM32 configuration and code-generation tooling. I turn functional requirements into technical solutions, work closely with hardware teams, and back my code with tests (ISTQB certified). Used to working as a consultant on client projects; looking for a long-term embedded role in Europe.

TECHNICAL SKILLS

Languages: C, C++ (C++17), embedded C, Java, Python
Embedded Linux: Linux kernel / device drivers, BSP, Android AOSP, HAL (Treble), SELinux, debugging on target
Microcontrollers: STM32 / ARM Cortex-M, ESP32, nRF52832, firmware, drivers, real-time constraints
Protocols: CAN, SPI, I2C, UART, Bluetooth / BLE, Wi-Fi, ESP-NOW
Design: requirements → technical specification, software architecture, design patterns, state machines, event-driven design, code reviews, root-cause analysis
Quality & testing: GoogleTest / GoogleMock, JUnit, regression testing, CI/CD, static analysis, ISTQB CTFL
Tools: Git, GitHub Actions, GitLab CI, Docker, SonarQube, Jira, Vector CANoe, ADB, Systrace, STM32CubeMX, STM32CubeIDE, Keil, EWARM

PROFESSIONAL EXPERIENCE

Embedded Software Engineer — STM32CubeMX

ACTIA Engineering Services · Client: STMicroelectronics · Tunis · Apr 2023 – present

- Developed the **Start from Board** feature: select a development board and instantly initialise its STM32 configuration.
- Contributed to **HSP integration and code generation for STM32U3** (ARM Cortex-M) microcontrollers.
- Building a **command-line interface (CLI)** for STM32CubeMX to enable automation and CI integration.
- Led **root-cause analysis** of complex defects and delivered lasting fixes.
- Acted as technical interface with Embedded Systems teams on hardware constraints, code generation and tool architecture.
- Designed, tested and integrated new features; took part in architecture reviews; set up JUnit regression tests and CI/CD support.

Environment: C/C++, Java, Python, STM32 / ARM Cortex-M, STM32CubeIDE, Keil, EWARM, Linux, Git, JUnit, CI/CD

Software Engineer — Android Automotive / Embedded Linux (Development & Validation)

ACTIA Engineering Services · Client: Renault · Tunis · Aug 2021 – Mar 2023

- Implemented **CAN firmware in embedded C at Linux kernel/driver level** (BSP) for the in-vehicle infotainment platform.
- Developed **C++17 system services** using design patterns and integrated custom **HALs** (Android Treble).
- Contributed to the **FOTA** (Firmware Over-The-Air) integration of the IVI platform (Alliance Automotive Software Platform).
- Debugged AOSP and fixed SELinux issues (ADB, Systrace); unit tests with GoogleTest/GoogleMock.
- Validated vehicle software updates (USB and server) and analysed CAN traces with Vector CANoe.

Environment: C/C++, embedded C, Linux kernel, Android/AOSP, CAN, Vector CANoe, Git, Jira

FINAL-YEAR ENGINEERING PROJECT

Embedded AI for High-Voltage Short-Circuit Detection

ACTIA Engineering Services · Client: EDF France · Feb 2021 – Aug 2021

- Wrote a **custom SPI driver** for the EVALSTPM33 sensor on STM32H745, as the standard ST driver could not meet the sampling constraints.
- Trained a neural network (ANN) on data from a purpose-built three-phase circuit.
- Deployed the model on STM32H745I-DISCO with STM32Cube.AI for **real-time detection** with on-board display.

Environment: STM32H745, embedded C, SPI, STM32Cube.AI, EVALSTPM33

PERSONAL PROJECTS

- **Distributed ESP-NOW weather monitoring** — ESP8266 sensor nodes + ESP32 gateway with an LVGL touchscreen UI (C++, ESP-NOW, DHT, OLED). *github.com/dalilawini/weather-app*
- **Custom embedded PCBs** — full hardware cycle: schematic and layout in Altium Designer, CNC fabrication (FlatCAM), assembly and bring-up testing.
- **Wireless prototypes** — Bluetooth HID and BLE on ESP32 and nRF52832, HM-10 / CC2541 and RF links.
- **JavaFX embedded GUI generator** — desktop tool that designs display screens and generates embedded-oriented UI code.

EDUCATION

- **Engineering Degree (MSc level) in Embedded Systems & Electronics** — Faculty of Sciences of Tunis (Univ. Tunis El Manar), 2021
- **Master's Degree in Electrical Engineering & Automation** — Faculty of Sciences of Tunis, 2020
- **Bachelor's Degree in Electrical Engineering & Automation** — Faculty of Sciences of Tunis, 2018

CERTIFICATION & LANGUAGES

ISTQB® Certified Tester Foundation Level (CTFL) v4.0 — 2026
Languages: English: professional working proficiency · French: fluent · Arabic: native